Seasonal Sales Analysis in Retail

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BATCH NO: 11

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REVIEW – 2 COMMENTS ADDRESSED

1. Modules of the project need to analyse
2. Tools /Requirements gather

ABSTRACT

Seasonal demand variation is one of the major challenges faced by retail businesses. Accurate forecasting of sales trends helps organizations improve inventory planning, reduce operational costs, and increase profitability. This project presents a web-based Seasonal Sales Forecasting System
that analyzes historical supermarket sales data and predicts future demand using the ARIMA time-series model. The application is developed using Python and Flask, allowing users to upload datasets, perform preprocessing, visualize trends, and generate forecasts through an interactive dashboard. The system combines statistical modeling and web technology to deliver a practical tool for data-driven decision-making in the retail sector.

INTRODUCTION

Retail sales are affected by seasonal factors like festivals, holidays, climate changes, and promotional campaigns.

- These factors cause fluctuations in customer demand.
- Because of these fluctuations, predicting future sales becomes difficult and complex.
- If businesses cannot predict seasonal trends properly, they may face stock shortages.
- Stock shortages lead to lost sales and unhappy customers.
- On the other hand, excess inventory leads to financial losses and increased storage costs.
- To manage these challenges, forecasting models are used.
- Forecasting models analyze historical sales data.
- They identify patterns, trends, and seasonal behaviour.
- Based on this analysis, they estimate future demand.
- This helps businesses plan inventory efficiently and reduce financial risks.

EXISTING SYSTEM

- Sales forecasting is mostly done **manually using spreadsheets**.
- Predictions are based on **basic analysis or experience**.
- **Time-consuming and less accurate**.
- Difficult to **identify seasonal trends**.
- No **automated forecasting or visualization tools**.

PROPOSED SYSTEM

- A **web-based Seasonal Sales Forecasting system** built using **Flask**.
- Users can **upload historical sales data (CSV)**.
- The system performs **data analysis and visualization**.
- Uses **ARIMA** to predict future sales.
- Provides **accurate forecasts and recommendations** for better inventory planning.

OBJECTIVES

The objectives of the proposed system include the following:-

1. To **analyze historical sales data**.
2. To **identify seasonal patterns and trends**.
3. To **predict future sales using forecasting models**.
4. To provide **data visualization and insights**.
5. To help businesses **improve inventory and sales planning**.

SOFTWARE REQUIREMENTS

Below are the software requirements for application development:

- **Programming Language:** Python
- **Web Framework:** Flask
- **Libraries:** Pandas, NumPy, Matplotlib, Statsmodels
- **IDE/Editor:** VS Code
- **Database:** CSV dataset
- **Web Browser:** Chrome

HARDWARE REQUIREMENTS

Hardware requirements for application development are as follows:

1. CPU - intel i3 or higher
2. RAM – 4 GB or higher
3. Storage – 10 GB or higher
4. OS - windows 10 or higher

FUNCTIONAL REQUIREMENTS

User Login: Secure login for authorized users.

Dataset Upload: Users can upload historical sales data (CSV).

Data Validation: Check uploaded data for errors or missing values.

Data Preprocessing: Clean and prepare data for analysis.

Exploratory Data Analysis (EDA): Generate graphs and charts to visualize trends.

Forecasting: Predict future sales using the **ARIMA model**.

Recommendations: Provide insights for inventory and sales planning.

Dashboard: Display all results and visualizations on an interactive web page.

NON - FUNCTIONAL REQUIREMENTS

Performance: Fast data processing and forecasting for smooth user experience.

Usability: Simple, user-friendly web interface for easy navigation.

Reliability: Accurate predictions and stable system performance.

Scalability: Can handle larger datasets as business grows.

Security: Protect user data and restrict access to authorized users only.

Maintainability: Easy to update code, models, and libraries.

Portability: Works on Windows, Linux, and MacOS platforms.

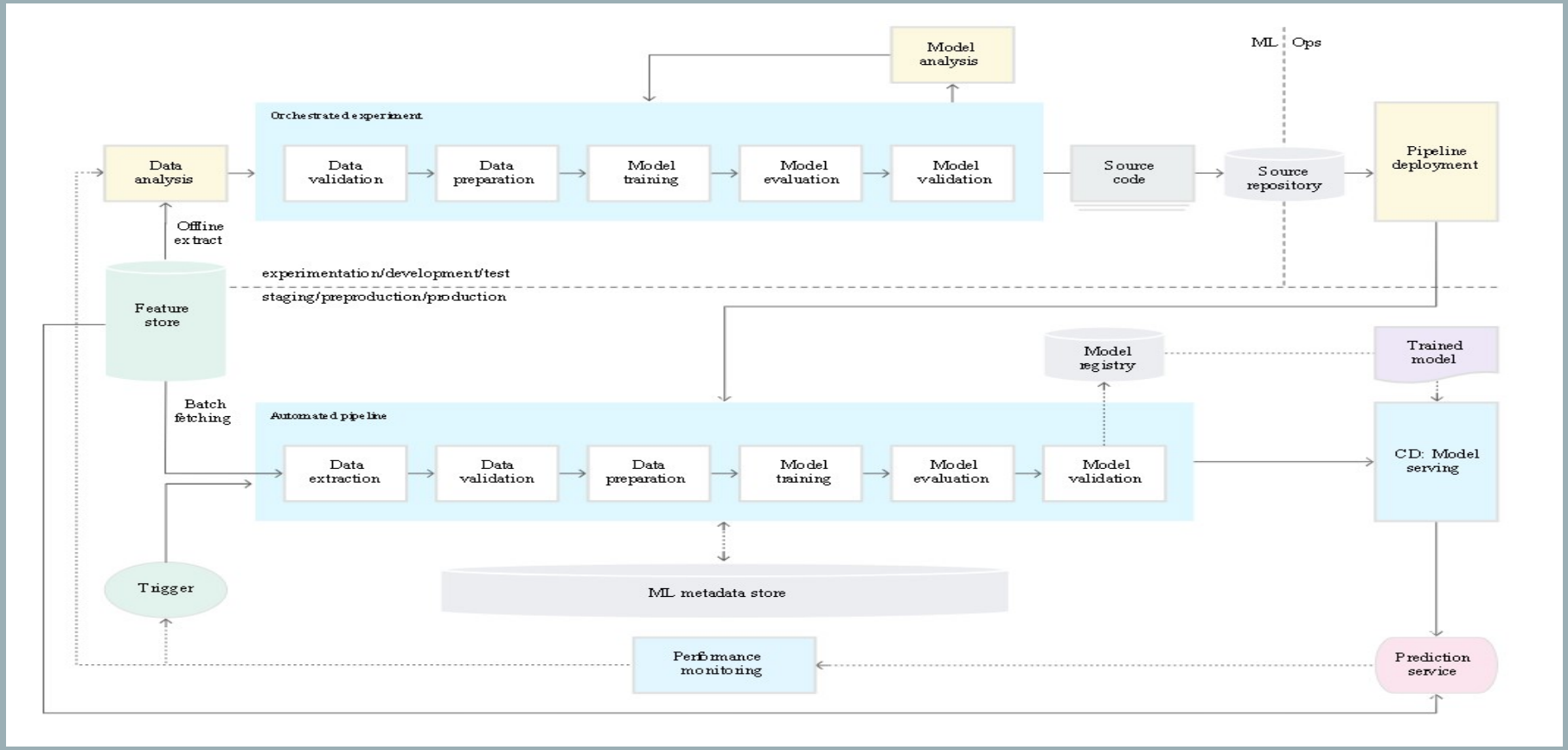
LITERATURE SURVEY

S. No	Name of the research paper	Author name	Technologies used	Advantages/ Disadvantages
1	Evaluation of Italian wine by the electronic tongue: recognition, quantitative analysis and corrolation with human sensory perception,2003	Legin, Rudnitskaya, Luvova, Vlosov, Natale, D'Amico	Array of sensors, Data processing techniques	Advantage: minimizes human risks involved Disadvantage: Time consuming and results affected by temperature
2	Selection of important features and predicting wine quality using ML techniques,2017	Yogesh Gupta	Linear regression, neural networks, SVM	Advantage: more precise predictions Disadvantage: results are not accurate for large datasets
3	MLOps Overview, Definition and Architecture,2021	Dominik Kreuzberger, Niklas Kuhl, Sebastian Hirschl	DevOps, ML pipeline	Advantage: automation of ML workflow Disadvantage: still need human assistance

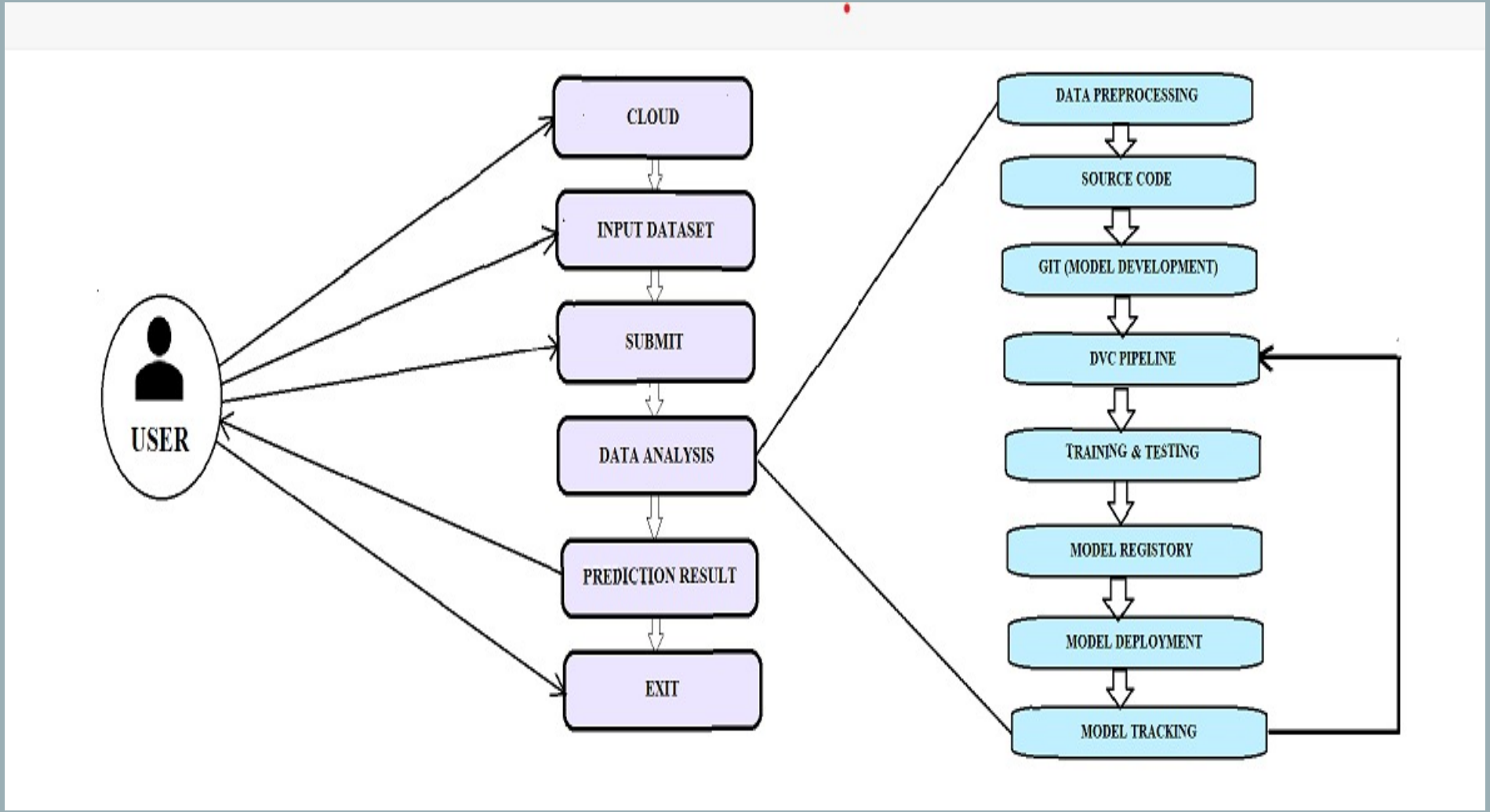
S. No	Name of the research paper	Author name	Technologies used	Advantages/ Disadvantages
4	The machine learning life cycle and the cloud: implications for drug discovery, 2021.	Ola Spjuth, Jens Fridb and Andreas Hellander	MLOps, Deep Neural Networks, Cloud computing.	Advantages: By using MLOps it gives support at all stages in the ML modeling lifecycle. Disadvantages: We need to build trust for collecting sensitive data from different sources.
5	A Data Quality-Driven View of MLOps, 2021. .	cedric.renggli, luka.rimanic, nezihe.guerel, bojan.karlas, ce.zhang.	CI/CD-Continuous integration and Continuous delivery, DevOps	Advantages: It improves the quality of the model. Disadvantages: Needs continuous monitoring.
6	Challenges in the deployment and operation of machine learning in practice,2020.	Lucas Baier, Fabian Jöhren Stefan Seebacher	CRISP-DM - standard process model for data analytics, conducted literature reviews and structured interviews with ML practitioners.	Advantage: pre – deployment, deployment, non - technical challenges are addressed clearly providing solutions. Disadvantage: could not cover all industry sectors, only limited statements can be made about the severity of one challenge in comparison to another.

SYSTEM ARCHITECTURE

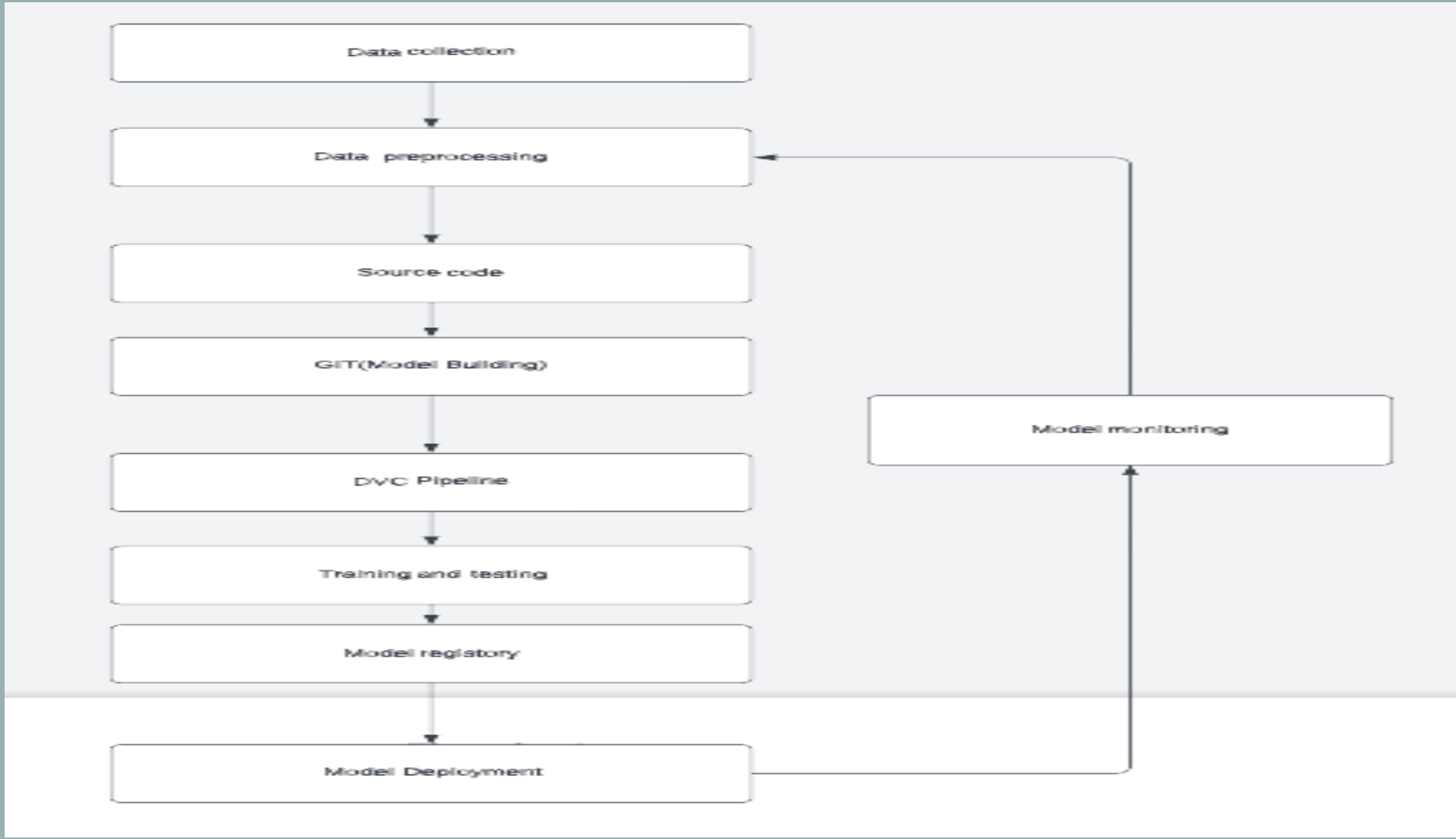
The goal is to perform continuous training of the model by automating the ML pipeline; this lets you achieve continuous delivery of model prediction service. To automate the process of using new data to retrain models in production, you need to introduce automated data and model validation steps to the pipeline, as well as pipeline triggers and metadata management.



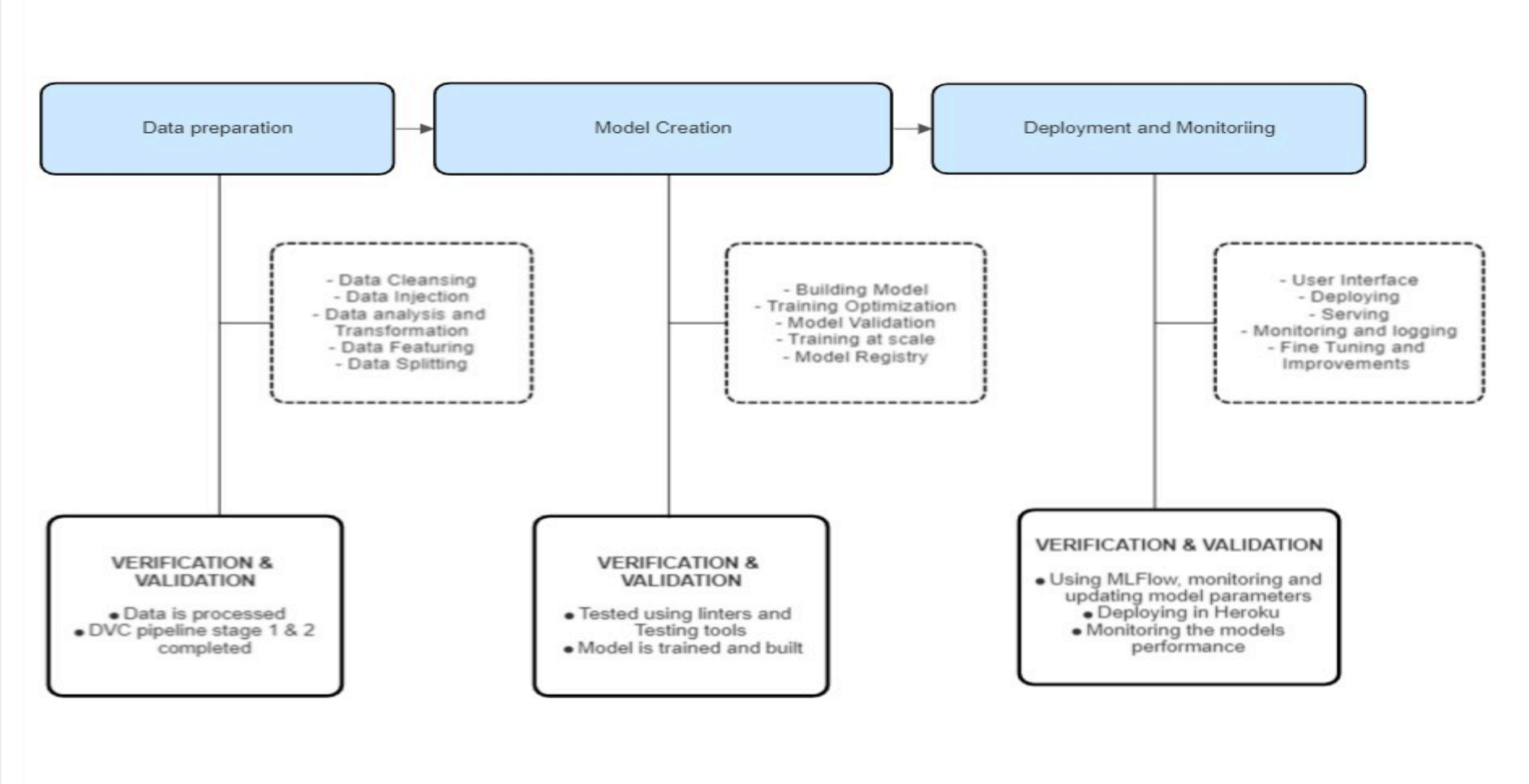
UML DIAGRAM


USER
CLOUD
INPUT DATASET
SUBMIT
DATA ANALYSIS
PREDICTION RESULT
EXIT
DATA PREPROCESSING
SOURCE CODE
GIT (MODEL DEVELOPMENT)
DVC PIPELINE
TRAINING & TESTING
MODEL REGISTORY
MODEL DEPLOYMENT
MODEL TRACKING

ACTIVITY DIAGRAM



MODULE DESIGN



THANK YOU